Software Maintenance and Neglect

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Types of Preventative Maintenance

- Corrective

  Replacing or fixing software after it fails

- Adaptive

  Modifying software to keep it operational after a change in its operating environment

- Perfective

  Adjusting software as necessary, adding or removing features as necessary.

The Cost of Not Performing Preventive Software Maintenance

Some potential consequences of not performing this maintenance are:

- Increased likelihood of system failures
- Decreased system performance
- Higher maintenance costs in the long term
- Negative impact on user experience

Ethical Considerations

Some ethical considerations must be made when running a service or product:

- Responsibility to users
- Responsibility to stakeholders
- Duty of care
- Ethical implications of negligence

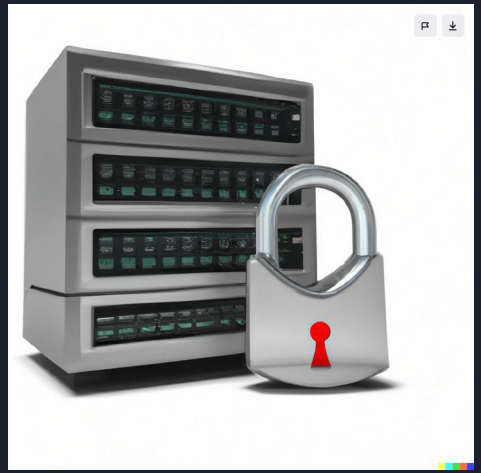
Real World Examples

- Equifax Data Breach (2017)
- TSB Bank IT Meltdown (2018)
- Boeing 737 MAX Crash (2019)

Mitigating the Risk of Not Performing Preventive Software Maintenance

Important and simple things that can mitigate risk:

- Implementing a preventive maintenance plan
- Regular software updates
- Investing in robust testing and quality assurance processes



Conclusion

What obligations, if any, does a software developer have to their consumer?

How can we ensure that companies are held accountable for the security and privacy of their customers' data in an increasingly complex and interconnected digital world?

Should companies be legally required to perform regular software maintenance and security audits to minimize the risk of data breaches and other cyber threats?

Are companies that prioritize their own profits over the security and privacy of their customers' data acting unethically, and if so, how can we encourage more responsible behavior?

How can we promote greater collaboration and information-sharing among companies, governments, and other stakeholders to improve overall cybersecurity?